Lecture Slide - 5

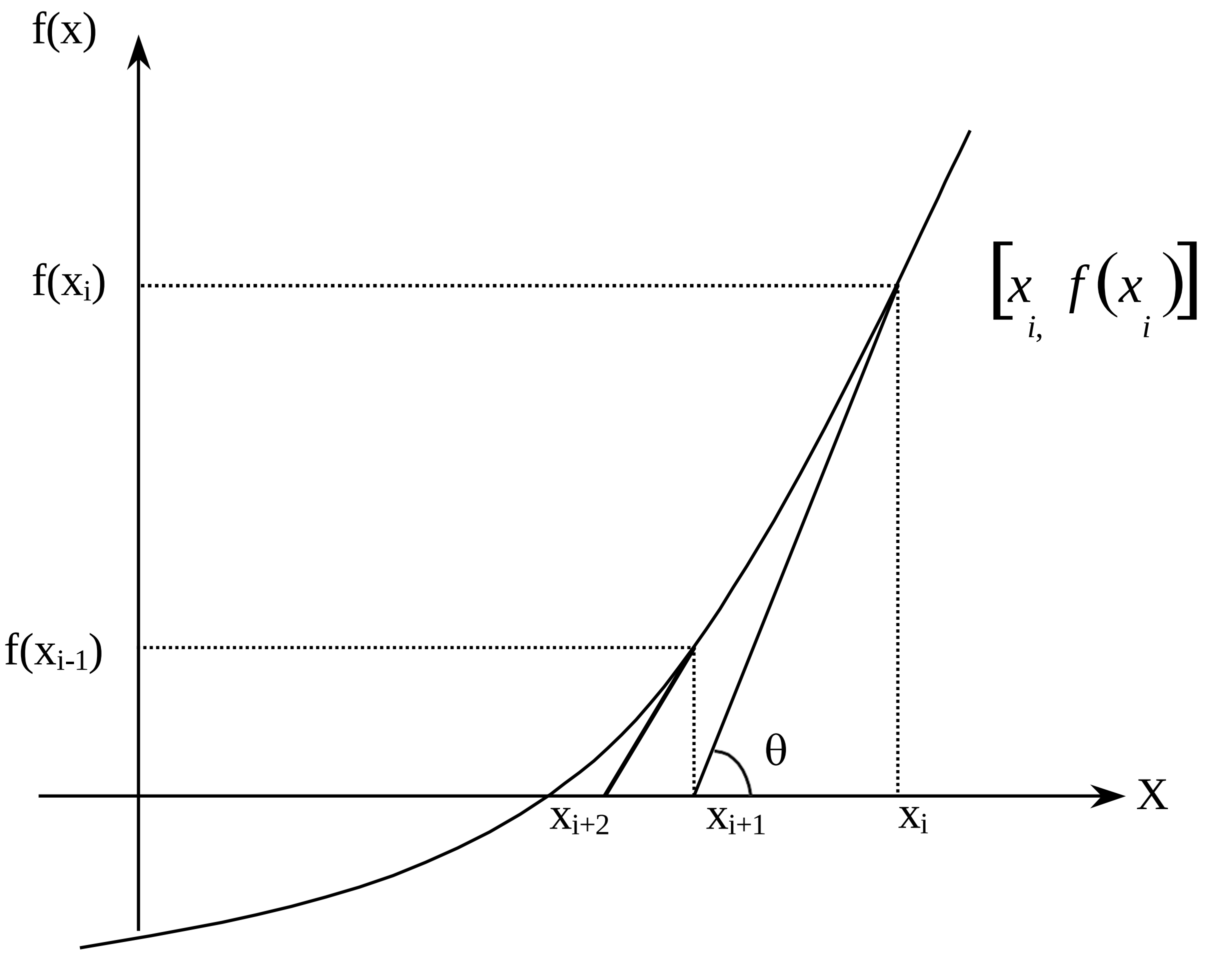
Open Methods -

Newton-Rapson & Secant Methods

Open Methods

- In bracketing methods, repeated application always result in closer estimates of the true value of the root.
- Such methods are said to be convergent because they move closer to the truth as the computation progress.
- In contrast, the open methods require only a single starting value or two starting values that do not necessarily bracket the root.
- As such, they sometimes diverge or move away from the true root as the computation progresses
- However, when the open methods converge they usually faster than the bracketing methods.

Newton-Raphson Method

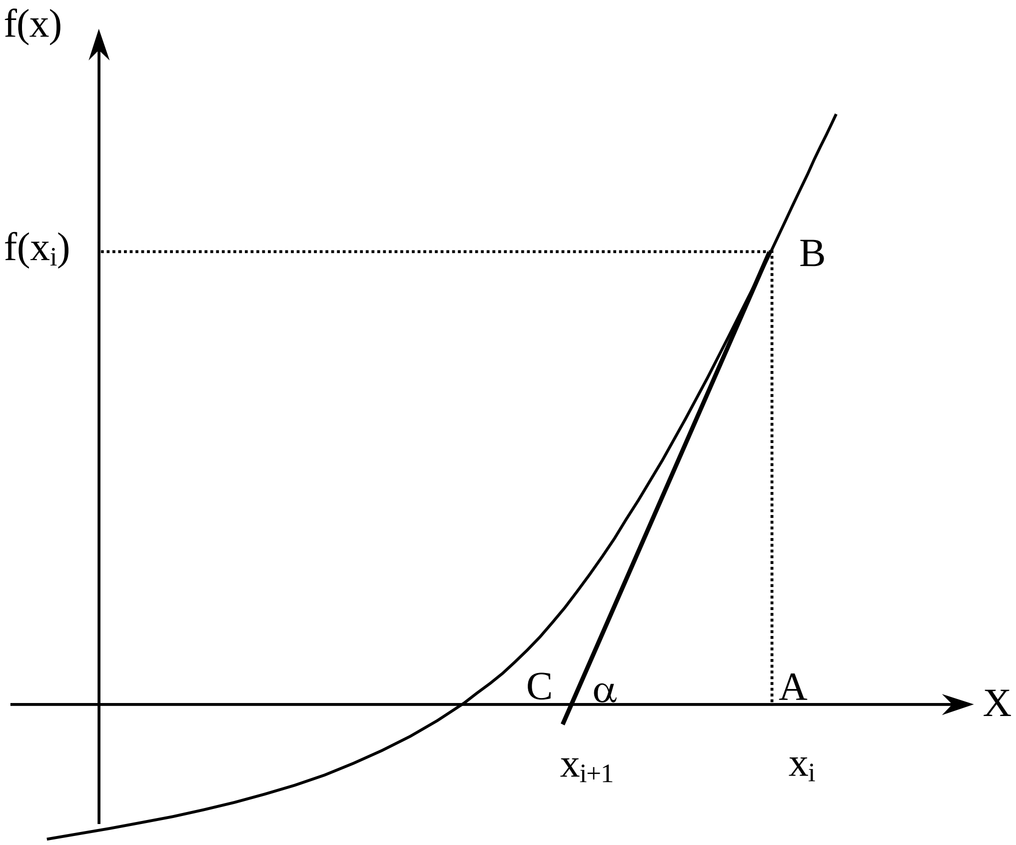


$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+2} = x_{i+1} - \frac{f(x_{i+1})}{f'(x_{i+1})}$$

Figure 1 Geometrical illustration of the Newton-Raphson method.

Derivation



$$\tan(\alpha) = \frac{AB}{AC}$$

$$f'(x_i) = \frac{f(x_i)}{x_i - x_{i+1}}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Figure 2 Derivation of the Newton-Raphson method.

Step 1

Example: Given, $f(x) = e^{-x} - x$,

Evaluate $f'(x)$ mathematically or symbolically,

$d(e^{-x} - x) = -e^{-x} - 1$

Step 2

Use an initial guess of the root, x_i, to estimate the new value of the root, x_{i+1} , as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

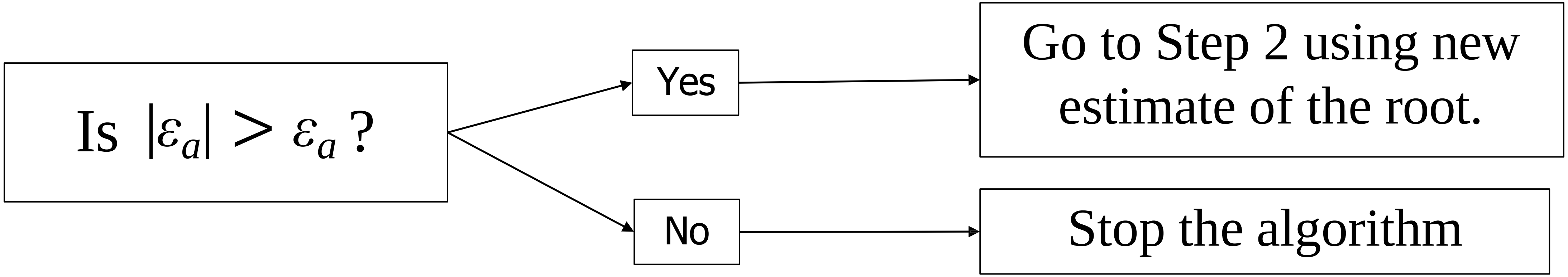
Step 3

Find the absolute relative approximate error $|\varepsilon_a|$ as;

$$|\varepsilon_a| = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| \times 100$$

Step 4

Compare the absolute relative approximate error with the pre-specified relative error tolerance ε_s



Also, check if the number of iterations has exceeded the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user.

Example 1

The floating ball has a specific gravity of *0.6* and has a radius of *5.5 cm*. You are asked to find the depth to which the ball is submerged when floating in water.

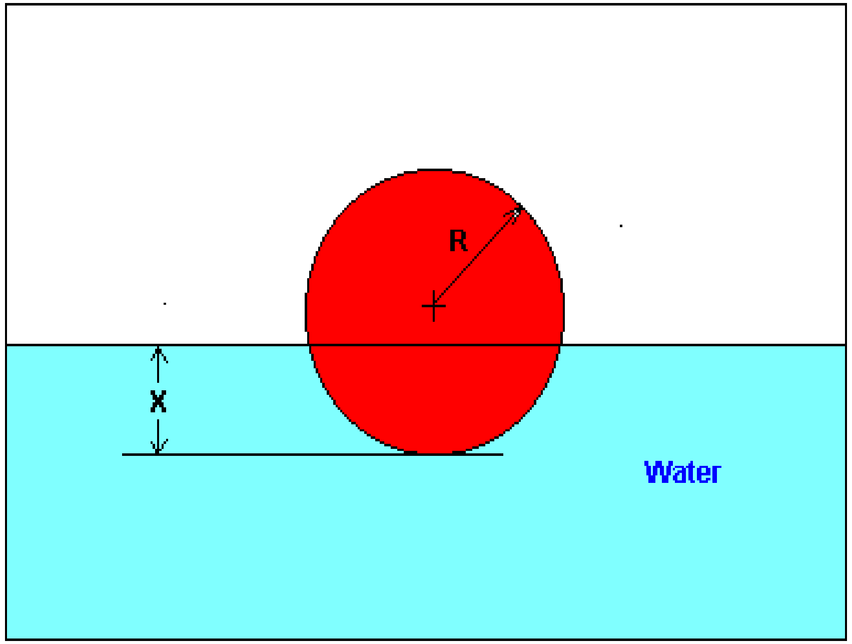


Figure 3 Floating ball problem.

Example 1 Cont.

The equation that gives the depth x in meters to which the ball is submerged under water is given by

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

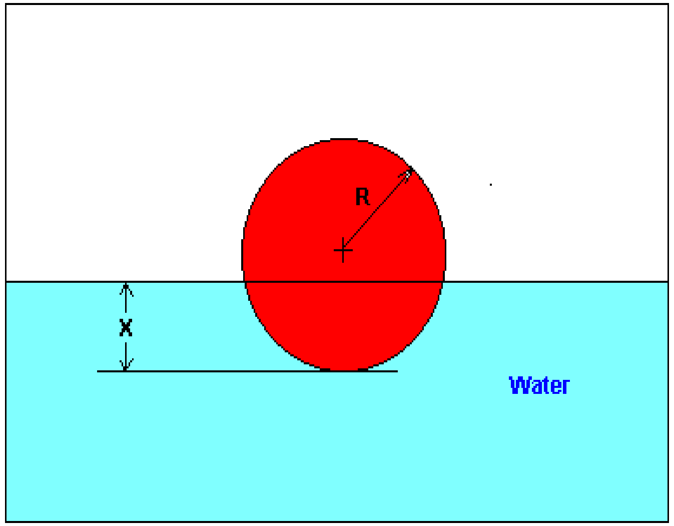


Figure 3 Floating ball problem.

Use the Newton-Raphson method of finding roots of equations to find

a) the depth 'x' to which the ball is submerged under water. Conduct three iterations to estimate the root of the above equation.

b) The absolute relative approximate error at the end of each iteration,

c) The number of significant digits at least correct at the end of each iteration.

Example 1 Cont.

Solution

To aid in the understanding of how this method works to find the root of an equation, the graph of *f(x)* is shown to the right,

where

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

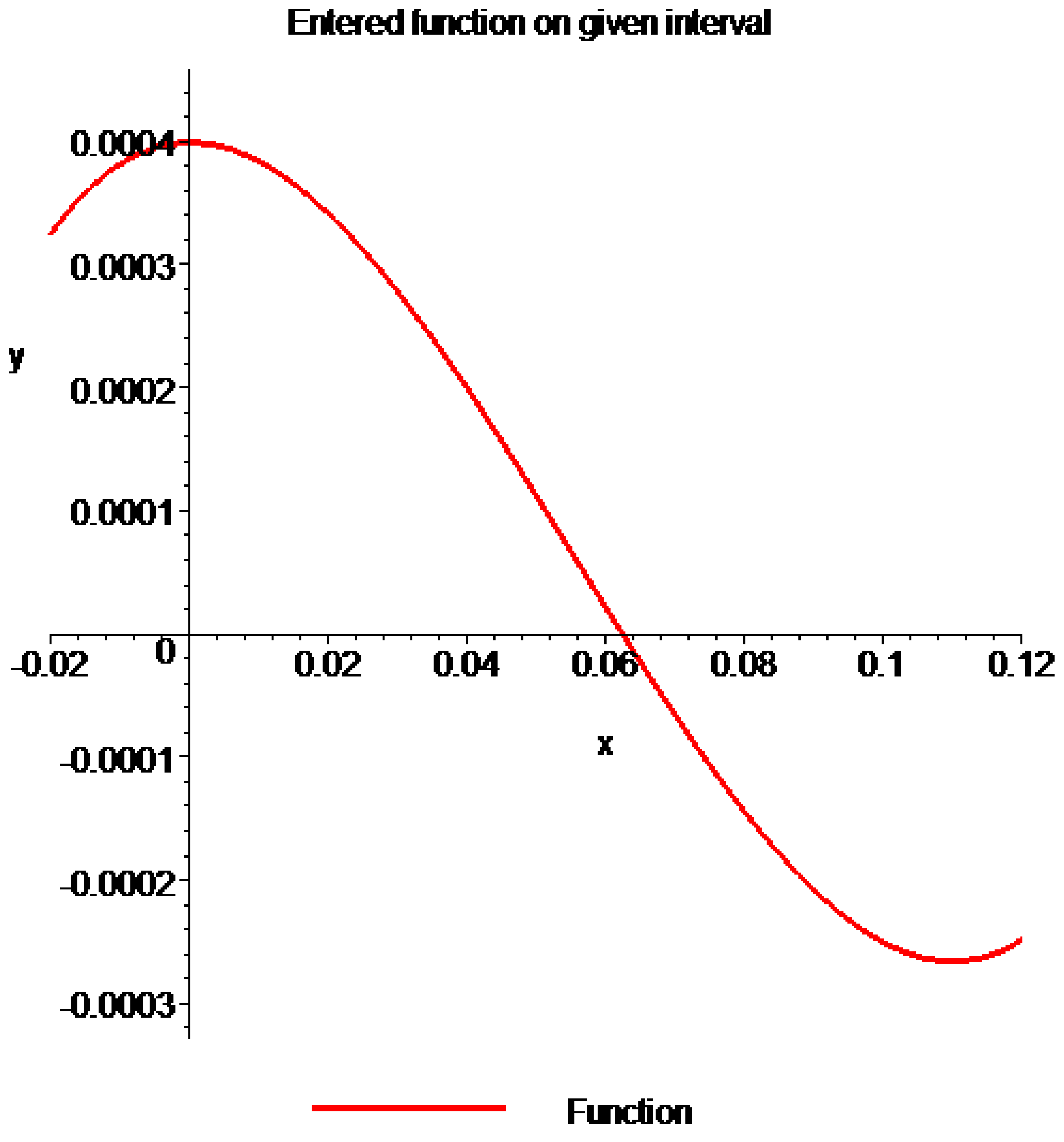


Figure 4 Graph of the function *f(x)*

Example 1 Cont.

Step-1: Evaluate $f'(x)$ mathematically or symbolically

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$
$$f'(x) = 3x^2 - 0.33x$$

Step-2: Let us assume the initial guess of the root of $f(x) = 0$ for $x_0 = 0.05$m. This is a reasonable guess (discuss why $x = 0$ and $x = 0.11$m are not good choices) as the extreme values of the depth x would be 0 and the diameter (0.11 m) of the ball.

Example 1 Cont. (Step-2)

Iteration 1

The estimate of the root is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993\times10^{-4}}{3(0.05)^2 - 0.33(0.05)}$$

$$= 0.05 - \frac{1.118\times10^{-4}}{-9\times10^{-3}}$$

$$= 0.05 - (-0.01242)$$

$$= 0.06242$$

Example 1 Cont.

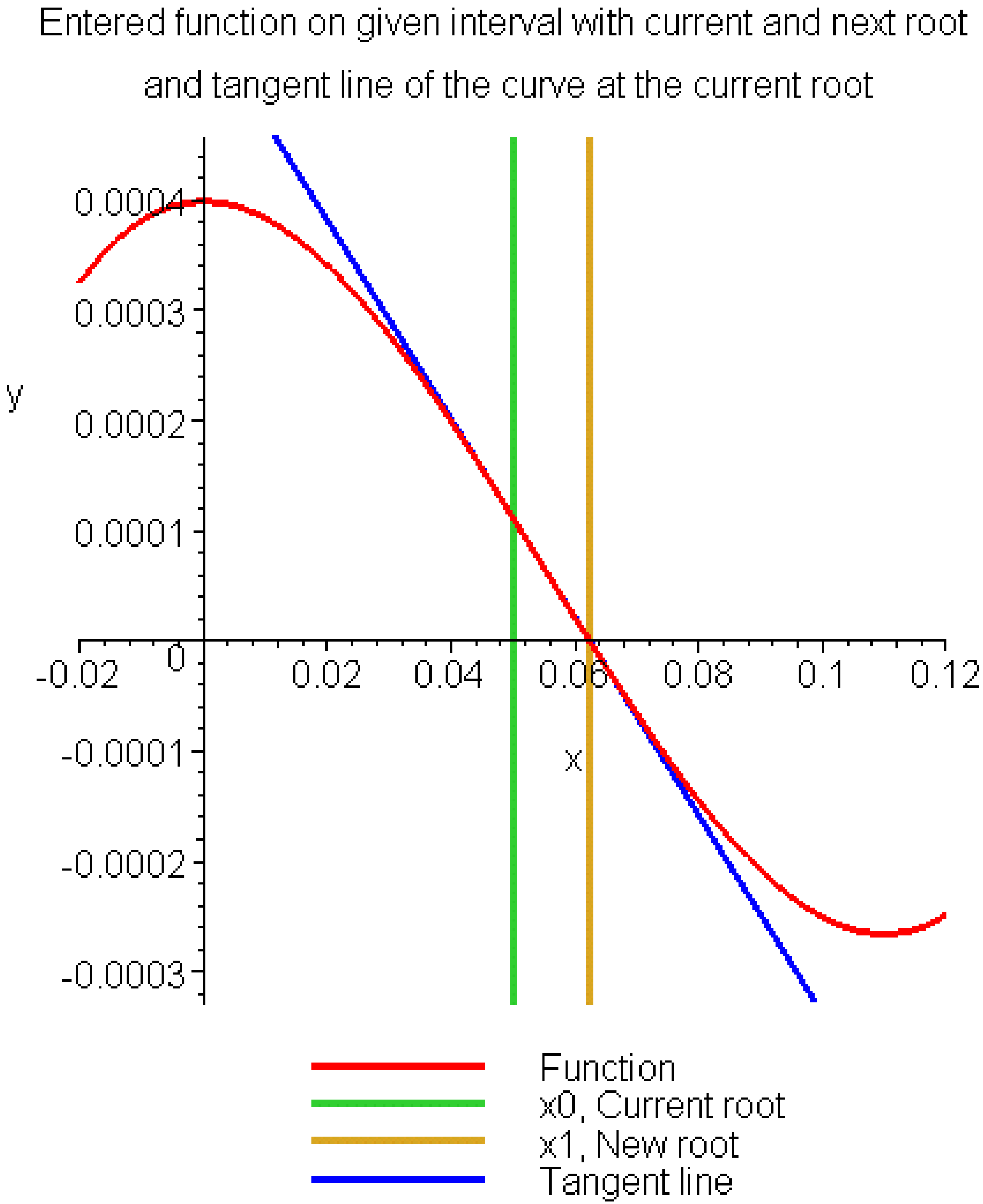


Figure 5 Estimate of the root for the first iteration.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 1 is

$$|\varepsilon_a| = \left|\frac{x_1 - x_0}{x_1}\right| \times 100$$

$$= \left|\frac{0.06242 - 0.05}{0.06242}\right|$$

$$= 19.90\%$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for at least one significant digits to be correct in your result.

Example 1 Cont.

Iteration 2

The estimate of the root is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993 \times 10^{-4}}{3(0.06242)^2 - 0.33(0.06242)}$$

$$= 0.06242 - \frac{-3.97781 \times 10^{-7}}{-8.90973 \times 10^{-3}}$$

$$= 0.06242 - (4.4646 \times 10^{-5})$$

$$= 0.06238$$

Example 1 Cont.

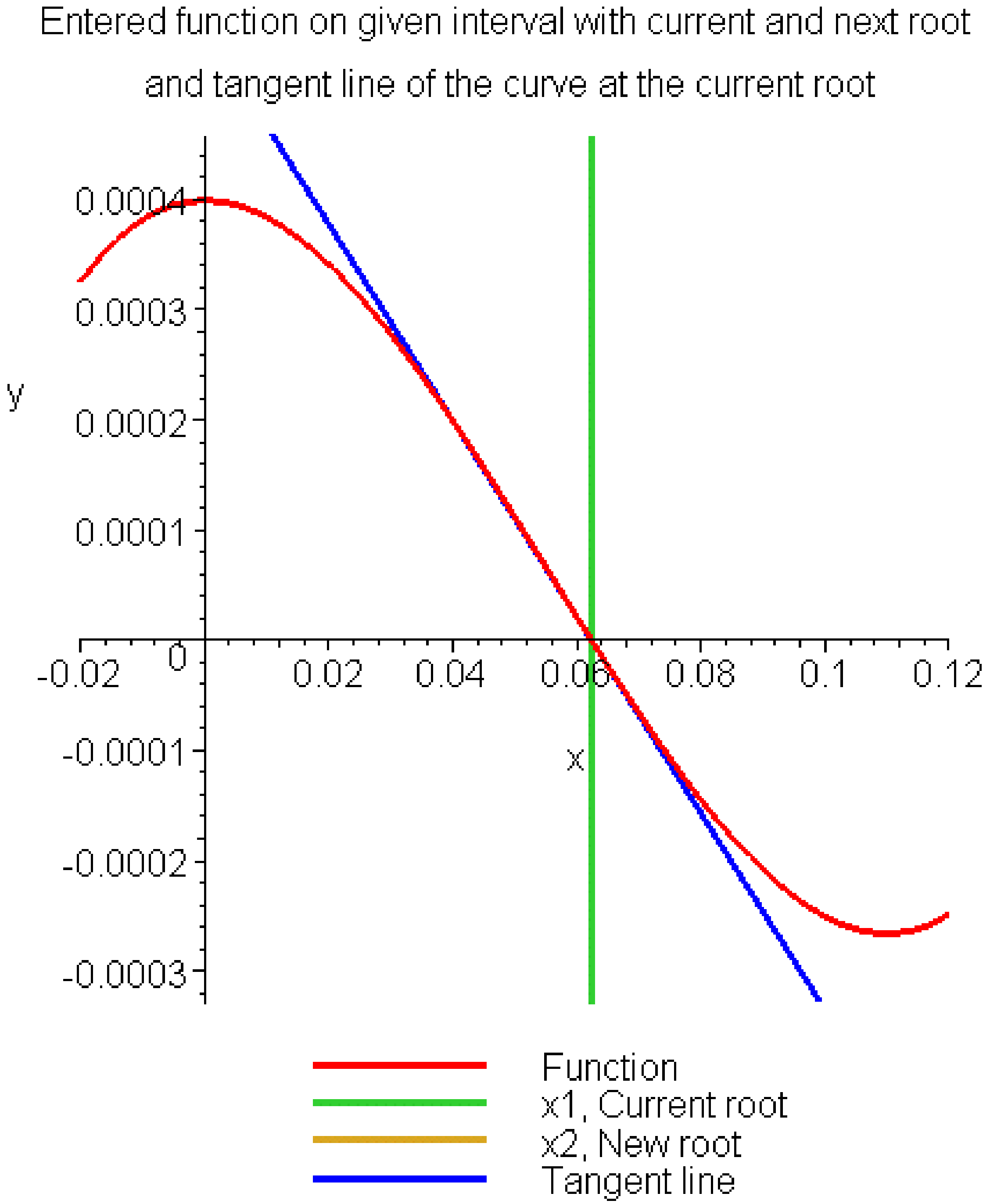


Figure 6 Estimate of the root for the Iteration 2.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 2 is

$$|\varepsilon_a| = \left|\frac{x_2 - x_1}{x_2}\right| \times 100$$

$$= \left|\frac{0.06238 - 0.06242}{0.06238}\right|$$

$$= 0.0716\%$$

The maximum value of m for which $|\epsilon_a| \leq 0.5 \times 10^{2-m}$ is 2.844. Hence, the number of significant digits at least correct in the answer is 2.

Example 1 Cont.

Iteration 3

The estimate of the root is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993\times10^{-4}}{3(0.06238)^2 - 0.33(0.06238)}$$

$$= 0.06238 - \frac{4.44\times10^{-11}}{-8.91171\times10^{-3}}$$

$$= 0.06238 - \left(-4.9822\times10^{-9}\right)$$

$$= 0.06238$$

Example 1 Cont.

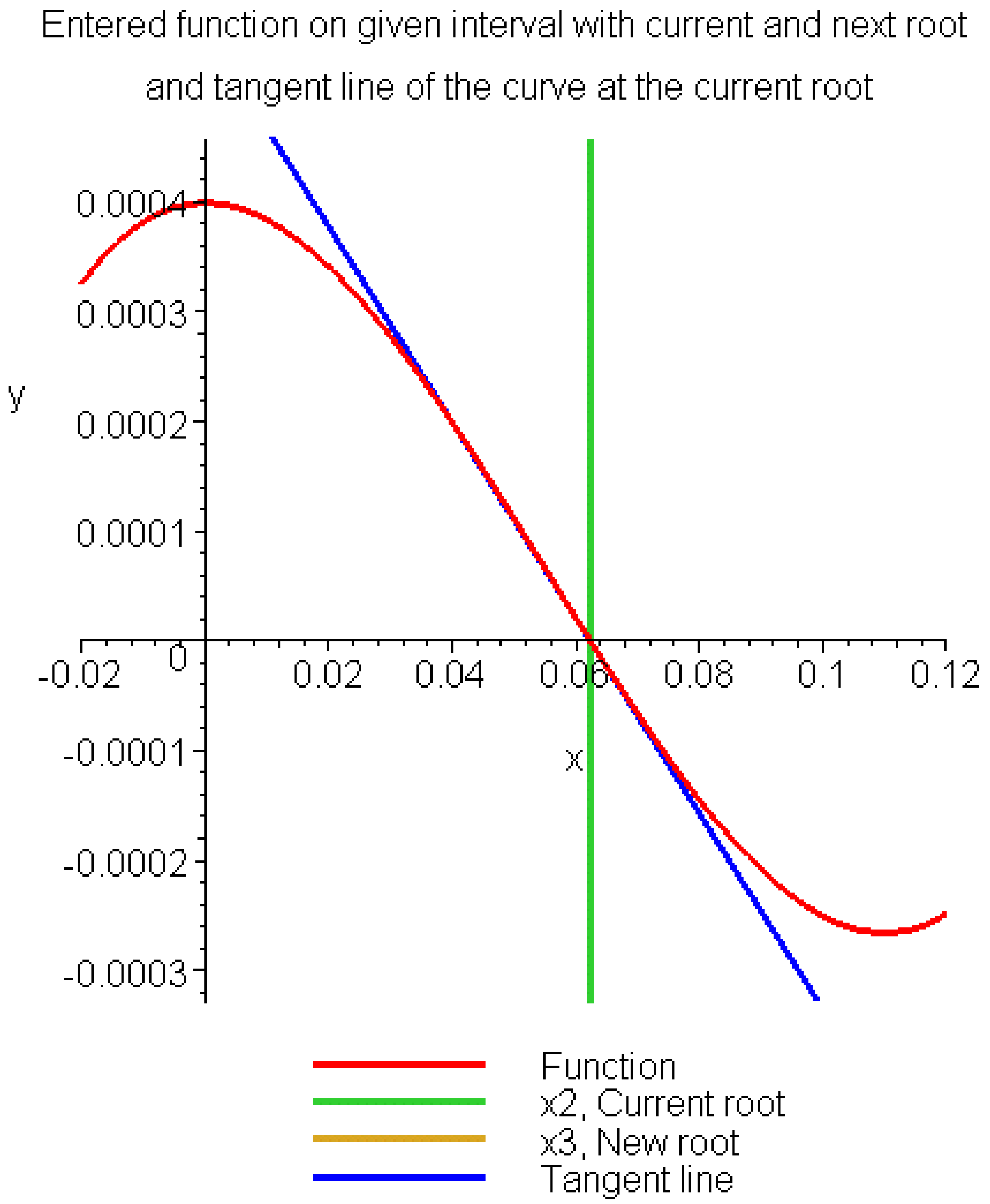


Figure 7 Estimate of the root for the Iteration 3.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 3 is

$$|\varepsilon_a| = \left|\frac{x_2 - x_1}{x_2}\right| \times 100$$

$$= \left|\frac{0.06238 - 0.06238}{0.06238}\right| \times 100$$

$$= 0\%$$

The number of significant digits at least correct is 4, as only 4 significant digits are carried through all the calculations.

Drawbacks of the Newton-Raphson Method

1. Divergence at inflection points

- If the initial guess or an iterated value turns out to be close to the inflection point of the function *f(x)* in the equation *f(x)=0*, the solution may start diverging away from the root.
- It may then start converging back to the root. For example, to find the root of the equation
- the Newton-Raphson's method reduces to

$$f(x) = (x-1)^3 + 0.512 = 0$$

$$x_{i+1} = x_i - \frac{(x_i - 1)^3 + 0.512}{3(x_i - 1)^2}$$

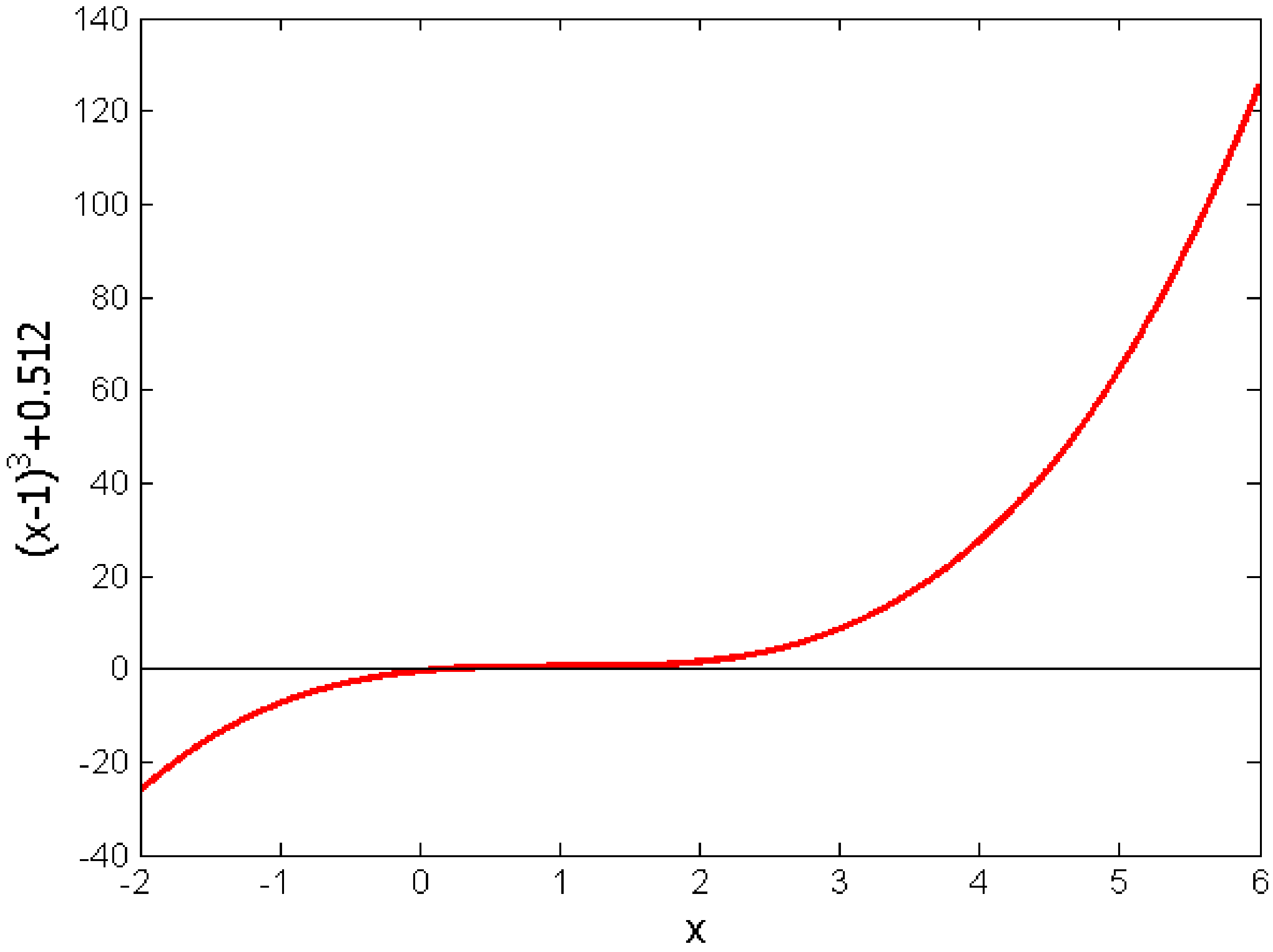
- Starting with an initial guess of x_0=5.0 , the next slide shows the iterated values of the root of the equation.

Table 1: Divergence near inflection point

- As you can observe, the root starts to diverge at Iteration 6 because the previous estimate of 0.92589 is close to the inflection point of $x=1$ (the value of $f'(x)$ is zero at the inflection point).
- Eventually, after 12 more iterations the root converges to the exact value of $x=0.2$

Iteration	x_i
0	5.0000
1	3.6560
2	2.7465
3	2.1084
4	1.6000
5	0.92589
6	–30.119
7	–19.746
8	–12.831
9	–8.2217
10	–5.1498
11	–3.1044
12	–1.7464
13	–0.85356
14	–0.28538
15	0.039784
16	0.17475
17	0.19924
18	0.2

Divergence at inflection point for $f(x) = (x-1)^3 + 0.512 = 0$



Drawbacks of the Newton-Raphson Method

2. Division by zero

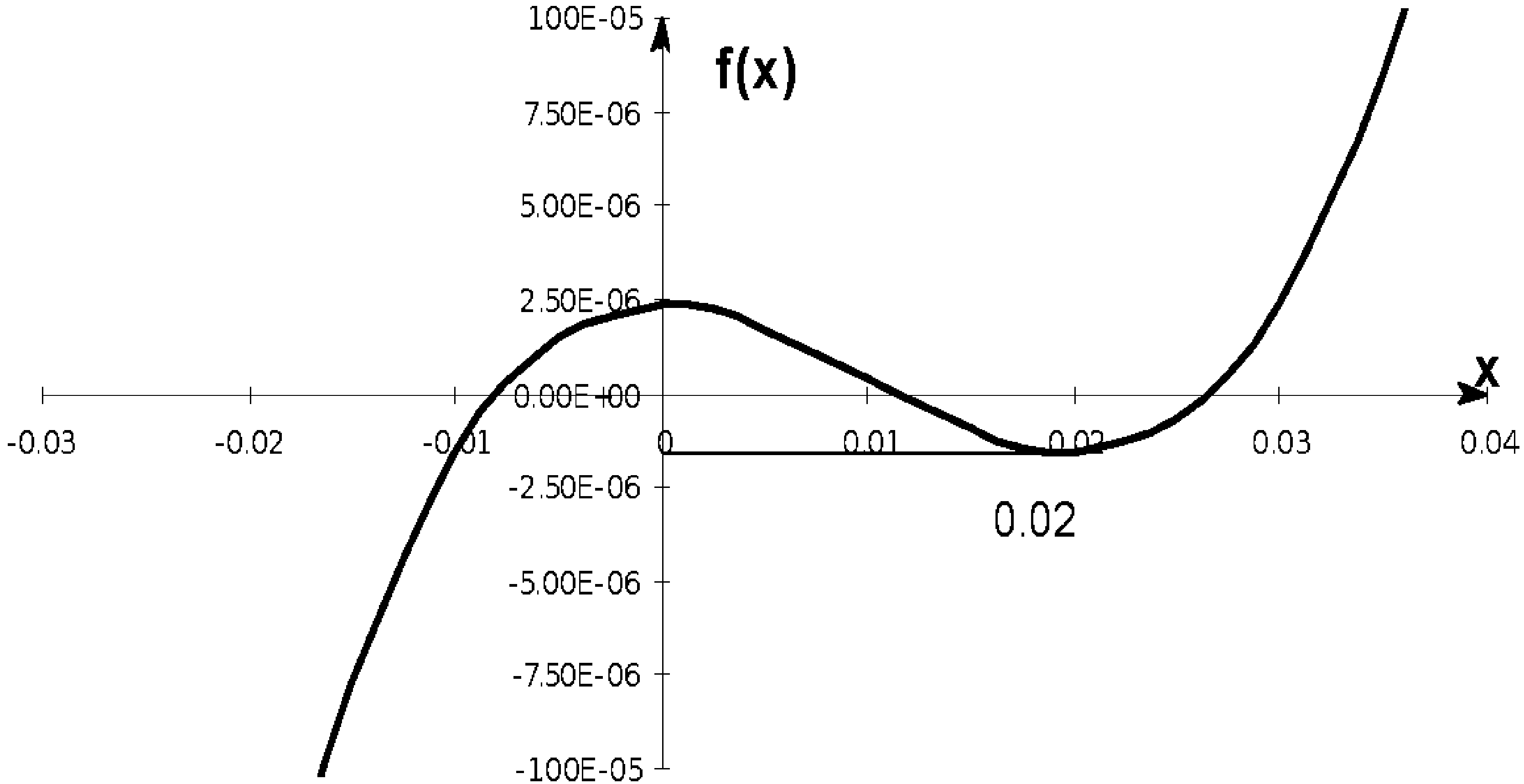
- For the equation $f(x) = x^3 - 0.03x^2 + 2.4\times10^{-6} = 0$

  the Newton-Raphson method reduces to

$$x_{i+1} = x_i - \frac{x_i^{\,3} - 0.03x_i^{\,2} + 2.4\times10^{-6}}{3x_i^{\,2} - 0.06x_i}$$

- For $x_i = 0$ or $x_i = 0.02$, division by zero occurs
- For an initial guess close to 0.02 such as x_0=0.019999, one may avoid division by zero, but then the denominator in the formula is a small number.
- For this case, as given in Table in the next slide, even after 9 iterations, the Newton-Raphson method does not converge.

Pitfall of division by zero or a near zero number



Division by near zero in Newton-Raphson method

Table 2

Iteration Number	x_i	$f(x_i)$	$\lvert \in_a \rvert \%$
0	0.019990		
1	–2.6480	18.778	100.75
2	–1.7620	–5.5638	50.282
3	–1.1714	–1.6485	50.422
4	–0.77765	–0.48842	50.632
5	–0.51518	–0.14470	50.946
6	–0.34025	–0.042862	51.413
7	–0.22369	–0.012692	52.107
8	–0.14608	–0.0037553	53.127
9	–0.094490	–0.0011091	54.602

Drawbacks of the Newton-Raphson Method

3. Oscillations near local maximum and minimum

- Results obtained from the Newton-Raphson method may oscillate about the local maximum or minimum without converging on a root but converging on the local maximum or minimum.
- Eventually, it may lead to division by a number close to zero and may diverge

For example, for $f(x) = x^2 + 2 = 0$

the equation has no real roots (Table 3).

Drawbacks of the Newton-Raphson Method

3. Oscillations near local maximum and minimum

For the given function:

$$f(x) = x^2 + 2$$

$$f'(x) = 2x$$

The iteration becomes:

$$x_{n+1} = x_n - \frac{x_n^2 + 2}{2x_n} = \frac{x_n}{2} - \frac{1}{x_n}$$

The method fails here due to two key properties of $f(x)$:

1. **No Real Roots:** The equation $f(x) = 0$ has no real solutions, as $x^2 + 2 = 0$ implies $x^2 = -2$.
2. **Existence of a Critical Point:** The derivative $f'(x) = 2x$ is zero at $x = 0$, which is a local minimum.

- If x_n is **positive**, $f(x_n) > 0$ and $f'(x_n) > 0$. The correction term $-\frac{f(x_n)}{f'(x_n)}$ is **negative**, so $x_{n+1} < x_n$.
- If x_n is **negative**, $f(x_n) > 0$ but $f'(x_n) < 0$. The correction term is **positive**, so $x_{n+1} > x_n$.

This causes the guesses to **oscillate** around $x = 0$. Furthermore, as $x_n \to 0$, $f'(x_n) \to 0$, making the step size $\left|\frac{f(x_n)}{f'(x_n)}\right| \to \infty$. This leads to large jumps and ultimate **divergence**.

Table 3: Oscillations near local maxima and minima in Newton-Raphson method

Iteration 0 ($n = 0$)

$$x_0 = -1.0000$$
$$f(x_0) = (-1)^2 + 2 = 3$$
$$f'(x_0) = 2 \times (-1) = -2$$
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = -1 - \frac{3}{-2} = -1 + 1.5 = \boxed{0.5}$$
$$|\epsilon_a|_1\% = \left|\frac{0.5 - (-1.0)}{0.5}\right| \times 100\% = \left|\frac{1.5}{0.5}\right| \times 100\% = \boxed{300.00\%}$$

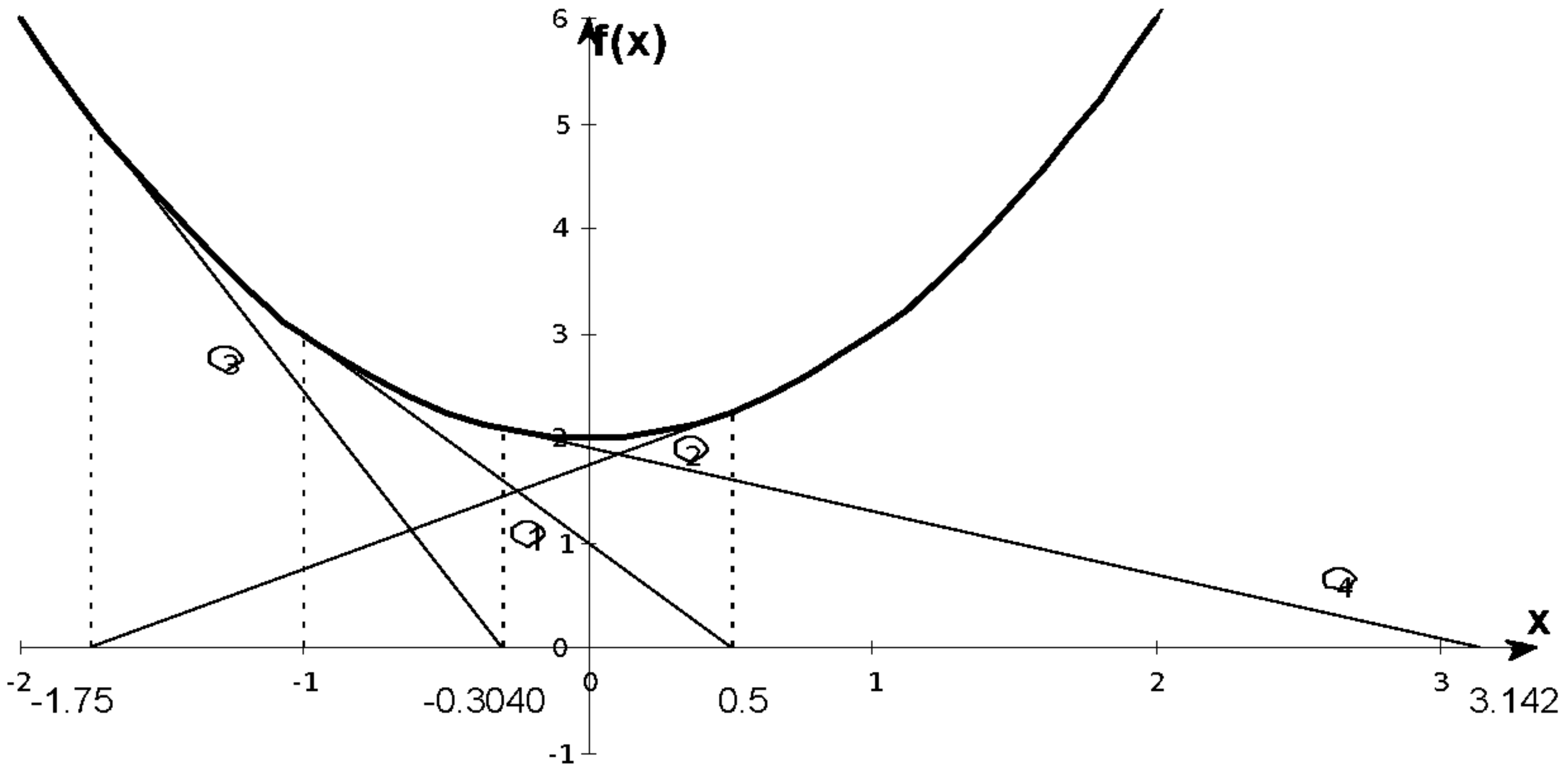
Iteration 1 ($n = 1$)

$$x_1 = 0.5$$
$$f(x_1) = (0.5)^2 + 2 = 2.25$$
$$f'(x_1) = 2 \times 0.5 = 1$$
$$x_2 = 0.5 - \frac{2.25}{1} = 0.5 - 2.25 = \boxed{-1.75}$$
$$|\epsilon_a|_2\% = \left|\frac{-1.75 - 0.5}{-1.75}\right| \times 100\% = \left|\frac{-2.25}{-1.75}\right| \times 100\% \approx \boxed{128.571\%}$$

Iteration Number		$f(x_i)$	$\lvert\in_a\rvert\%$
0	–1.0000	3.00	300.00
1	0.5	2.25	128.57
2	–1.75	5.063	1
3	–0.30357	2.092	476.47
4	3.1423	11.874	109.66
5	1.2529	3.570	150.80
6	–0.17166	2.029	829.88
7	5.7395	34.942	102.99
8	2.6955	9.266	112.93
9	0.97678	2.954	175.96

Oscillations around local minima for

$$f(x) = x^2 + 2$$



Drawbacks of the Newton-Raphson Method

4. Root jumping

- In some case where the function *f(x)* is oscillating and has a number of roots, one may choose an initial guess close to a root.
- However, the guesses may jump and converge to some other root.
- For example for solving the equation $\sin x=0$ if you choose $x_0 = 2.4\pi = (7.539822)$ as an initial guess, it converges to the root of $x=0$ as shown in Table 4.
- However, one may have chosen this as an initial guess to converge to

$$x = 2\pi = 6.2831853$$

Table 4: Root jumping in Newton-Raphson method

- **Iteration 0 ($n = 0$):**

$$x_0 = 2.4\pi \approx 7.539822$$
$$f(x_0) = \sin(2.4\pi) \approx 0.951$$
$$f'(x_0) = \cos(2.4\pi) \approx 0.309$$
$$\Delta x_0 = -\frac{f(x_0)}{f'(x_0)} \approx -\frac{0.951}{0.309} \approx -3.077$$
$$x_1 = x_0 + \Delta x_0 \approx 7.539822 - 3.077 \approx 4.462$$

- **Iteration 1 ($n = 1$):**

$$x_1 \approx 4.462$$
$$f(x_1) = \sin(4.462) \approx -0.969$$
$$f'(x_1) = \cos(4.462) \approx -0.247$$
$$\Delta x_1 = -\frac{f(x_1)}{f'(x_1)} \approx -\frac{-0.969}{-0.247} \approx -3.922$$
$$x_2 = x_1 + \Delta x_1 \approx 4.462 - 3.922 \approx 0.540$$

Iteration Number	x_i	$f(x_i)$	$\left\|\epsilon_a\right\|\%$
0			
1	7.539822	0.951	
2	4.462	–0.969	68.973
3	0.5499	0.5226	711.44
4	–0.06307	–0.06303	971.91
5			

Root jumping in Newton-Raphson method

solve the problem $f(x) = \sin(x) = 0$ using the Newton-Raphson method with the intended initial guess $x_0 = 2\pi$.

- **Initial Guess:** $x_0 = 2\pi$
- **Exact Target Root:** $x^* = 2\pi$

Iteration 0 (n = 0)

$$x_0 = 2\pi \approx 6.283185307179586$$

$$f(x_0) = \sin(2\pi) = 0$$

$$f'(x_0) = \cos(2\pi) = 1$$

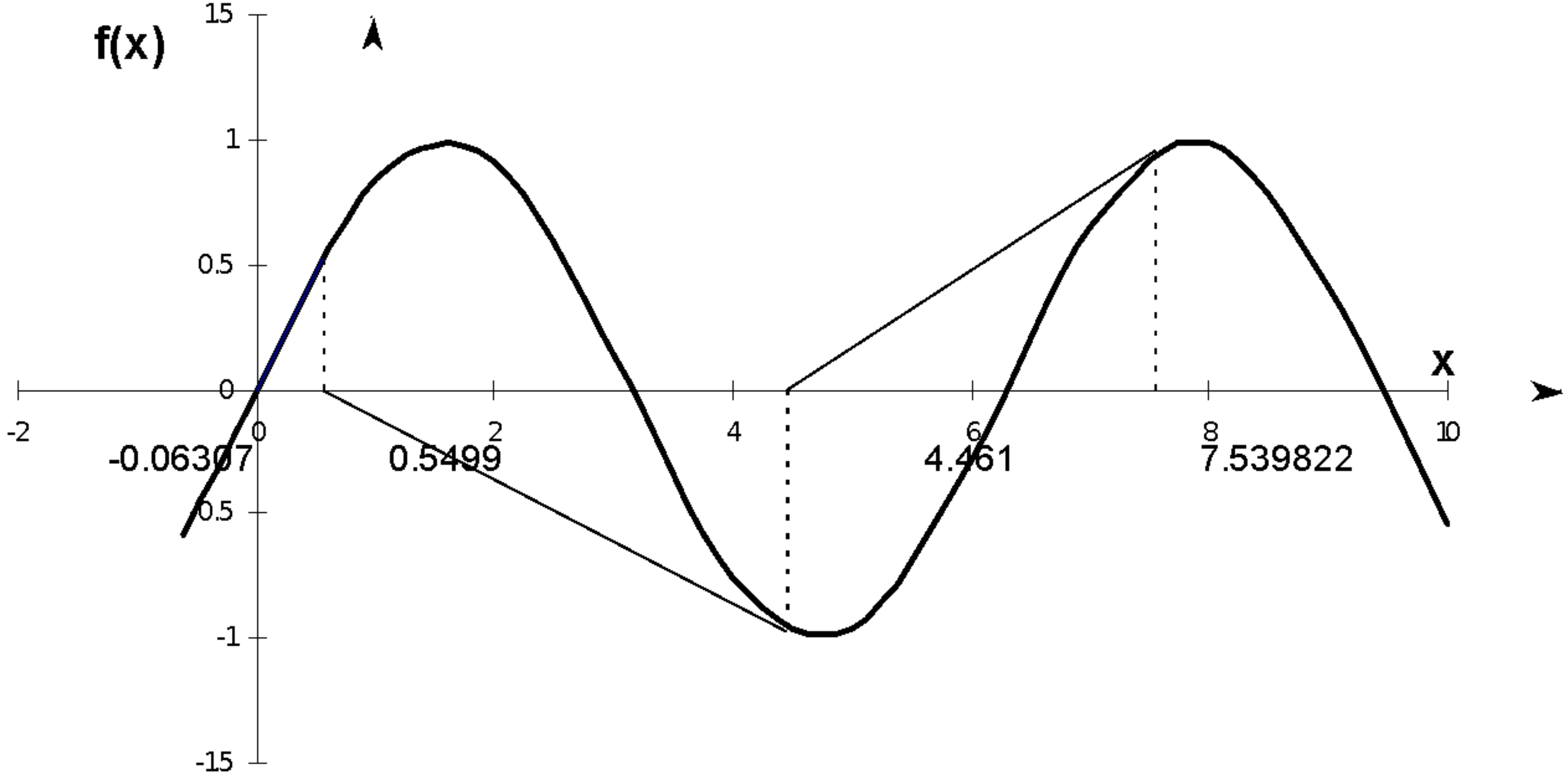
$$\Delta x_0 = -\frac{f(x_0)}{f'(x_0)} = -\frac{0}{1} = 0$$

$$x_1 = x_0 + \Delta x_0 = 2\pi + 0 = 2\pi$$

Observation: Since $f(x_0) = 0$, the algorithm has **immediately found the exact root** on the first iteration. The relative error is 0%, and the process terminates.

Contrast with the Previous Example ($x_0 = 2.4\pi$): This highlights the extreme sensitivity of the Newton-Raphson method to the initial guess. A slight perturbation from 2π to 2.4π caused a massive "root jump" and convergence to a different root ($x = 0$), while the exact guess led to immediate convergence.

Figure 6 Root jumping from intended location of root for $f(x) = \sin x = 0$



Key Takeaways

- As with other root-location methods, Eq. (3) can be used as a termination criterion.
- In addition, a theoretical analysis (Chapra and Canale, 2010) provides insight regarding the rate of convergence as expressed by

$$\varepsilon_{t,i+1} = -\frac{f''(x_r)}{2f'(x_r)}\varepsilon_{t,i}^2$$

- Thus, the error should be roughly proportional to the square of the previous error. In other words, the number of significant figures of accuracy approximately doubles with each iteration. This behavior is called quadratic convergence and is one of the major reasons for the popularity of the method.

The Secant Method

- A slight variation of Newton's method for functions whose derivatives are difficult to evaluate. $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$
- For these cases the derivative can be approximated by a backward finite divided difference.

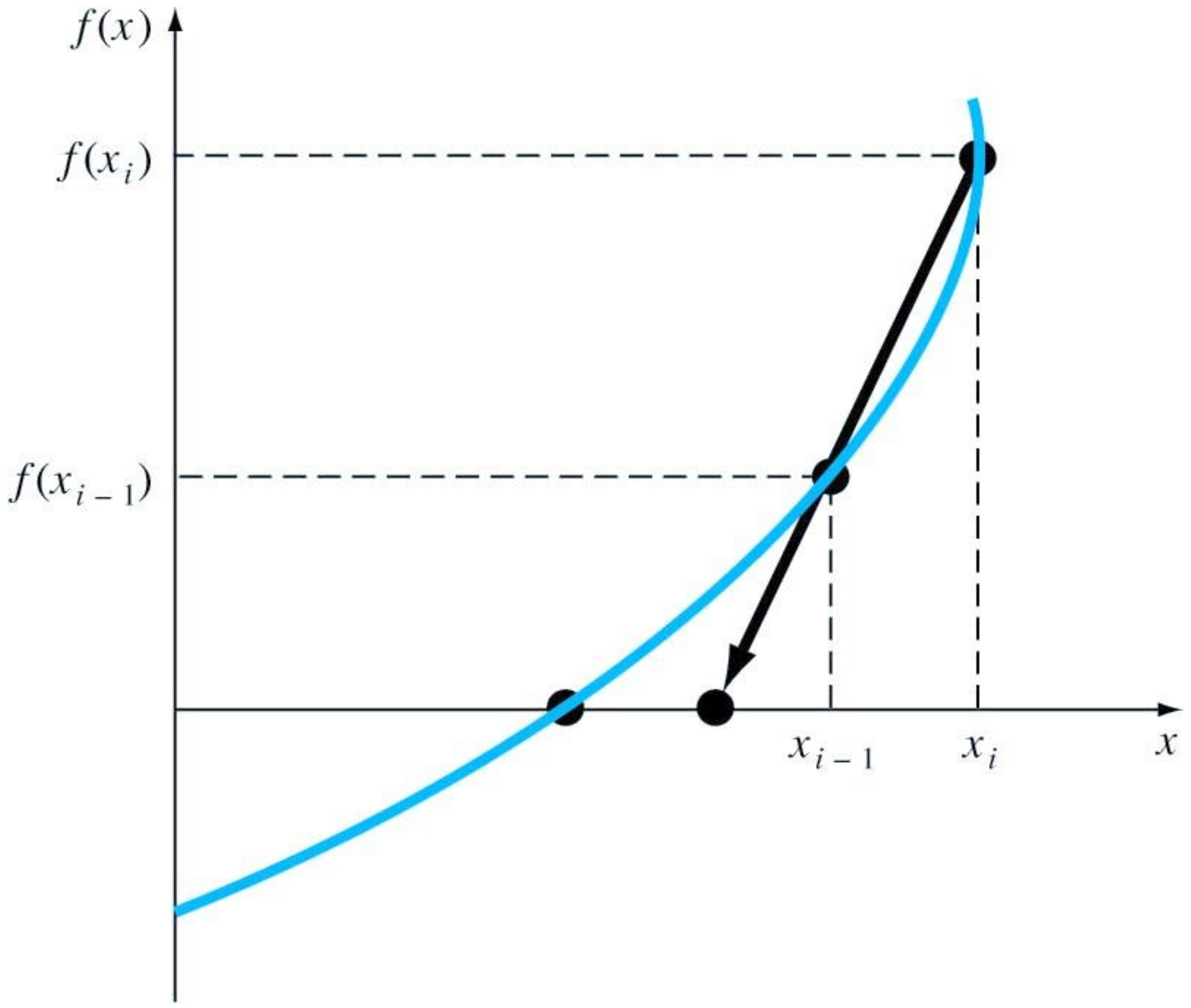
$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \qquad \frac{1}{f'(x_i)} \cong \frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})}$$

$$x_{i+1} = x_i - f(x_i)\frac{x_i - x_{i-1}}{f(x_i) - f(x_{i-1})} \quad i = 1,2,3,\ldots$$

- To estimate x_{i+1}, it needs $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$

The Secant Method

- Requires two initial estimates of x, *e.g.*, x_0, x_1. However, because $f(x)$ is not required to change signs between estimates, it is not classified as a “bracketing” method.
- The secant method has the same properties as Newton’s method. Convergence is not guaranteed for all x_0 *and* x_1.



Algorithm

Secant method Steps (Rule),

Step-1: Find points x_0 and x_1 such that $x_0<x_1$ and $f(x_0)\cdot f(x_1)<0$.

Step-2: Find next value using

Formula-1 : $x_2 = x_0 - f(x_0)\dfrac{x_1 - x_0}{f(x_1) - f(x_0)}$

or Formula-2 : $x_2 = \dfrac{x_0 \cdot f(x_1) - x_1 \cdot f(x_0)}{f(x_1) - f(x_0)}$

or Formula-3 : $x_2 = x_1 - f(x_1)\dfrac{x_1 - x_0}{f(x_1) - f(x_0)}$

(Using any of the formula, we will get same x_2 value)

Algorithm

Step-3: If $f(x_2)=0$ *then* x_2 is an exact root,

else $x_0 = x_1$ and $x_1 = x_2$

Step-4: Repeat steps 2 & 3 until $f(x_i)=0$ or

$|f(x_i)| \leq$ Accuracy

Example,

Find a root of an equation $f(x) = x^3 - x - 1$ using Secant method

Solution: List of some x and $f(x)$ values :

x	0	1	2
$f(x)$	-1	-1	5

1st iteration: Assume $x_0=1$ and $x_1=2$

Then, $f(x0) = f(1) = -1$ and $f(x1) = f(2) = 5$

$$\therefore x_2 = x_0 - f(x_0)\frac{x_1 - x_0}{f(x_1) - f(x_0)}$$

$x_2 = 1 - (-1)\times(2 - 1)/(5 - (-1))$

$x_2 = 1.16667$

$\therefore f(x_2) = f(1.16667) = -0.5787$

Example,

2nd iteration : $x_1=2$ and $x_2=1.16667$

$f(x_1) = f(2) = 5$ and $f(x_2) = f(1.16667) = -0.5787$

$$\therefore x_3 = x_1 - f(x_1)\frac{x_2 - x_1}{f(x_2) - f(x_1)}$$

$x_3=2 - 5\times(1.16667-2)/(-0.5787-5)$

$x_3=1.25311$

$\therefore f(x_3)=f(1.25311)=-0.28536$

Example,

<u>3^{rd} iteration</u> : x_2=1.16667 and x_3=1.25311

$f(x_2)=f(1.16667)=-0.5787$

and $f(x_3)=f(1.25311)=-0.28536$

$$\therefore x_4 = x_2 - f(x_2)\frac{x_3-x_2}{f(x_3)-f(x_2)}$$

$x_4=1.16667-(-0.5787)\times(1.25311-1.16667)/$

$(-0.28536-(-0.5787))$

$x_4=1.33721$

$\therefore f(x_4)=f(1.33721)=0.05388$

Example,

5th iteration : x_4=1.33721 and x_5=1.32385

$f(x_4)$=f(1.33721)=0.05388 and

$f(x_5)$=f(1.32385)=-0.0037

$$\therefore x_6 = x_4 - f(x_4)\frac{x_5 - x_4}{f(x_5) - f(x_4)}$$

x_6=1.33721-0.05388×(1.32385-1.33721)/(-0.0037-0.05388)

x_6=1.32471

∴ $f(x_6)$=f(1.32471)=-0.00004

Apply Secant Method in the floating ball problem

$$f(x) = x^3 - 0.165x^2 + 3.993\times10^{-4}$$

- Let us assume the initial guesses of the root of f(x)=0 as x_1=0.02 and x_0=0.05

$$x_1 = x_0 - \frac{f(x_0)(x_0 - x_{-1})}{f(x_0) - f(x_{-1})}$$

$$= x_0 - \frac{\left(x_0^3 - 0.165x_0^2 + 3.993\times10^{-4}\right)\times\left(x_0 - x_{-1}\right)}{\left(x_0^3 - 0.165x_0^2 + 3.993\times10^{-4}\right) - \left(x_{-1}^3 - 0.165x_{-1}^2 + 3.993\times10^{-4}\right)}$$

$$= 0.05 - \frac{\left[0.05^3 - 0.165(0.05)^2 + 3.993\times10^{-4}\right]\times\left[0.05 - 0.02\right]}{\left[0.05^3 - 0.165(0.05)^2 + 3.993\times10^{-4}\right] - \left[0.02^3 - 0.165(0.02)^2 + 3.993\times10^{-4}\right]}$$

$$= 0.06461$$

Secant Method: Floating ball problem (continued)

- The absolute relative approximate error $\left|\in_a\right|$ at the end of Iteration 1 is

$$\left|\in_a\right| = \left|\frac{x_1 - x_0}{x_1}\right| \times 100 = 22.62\%$$

- As you need an absolute relative approximate error of 5% or less so you need more iteration to carry on.

Iteration 2

$$x_2 = x_1 - \frac{f(x_1)(x_1 - x_0)}{f(x_1) - f(x_0)} = 0.06241$$

- The absolute relative approximate error $\left|\in_a\right|$ at the end of Iteration 2 is 3.525%

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Secant Method: Floating ball problem (continued)

- X_3 = 0.06238
- The absolute relative approximate error at the end of Iteration 3 is 0.0595%
- Table 1 shows the secant method calculations for the results from the above problem.

Table 1: Secent Method Result as Function of Iteration

Iteration Number, I	x_{i-1}	x_i	x_{i+1}	$\lvert\in_a\rvert\%$	$f(x_{i+1})$
1	0.02	0.05	0.06461		$-1.9812 \mathrm{X} 10^{-5}$
2	0.05	0.06461	0.06241	22.62	$-3.2852 \mathrm{X} 10^{-7}$
3	0.06461	0.06241	0.06238	3.525	$2.0252 \mathrm{X} 10^{-9}$
4	0.06241	0.06238	0.06238	0.0595	$-1.8576 \mathrm{X} 10^{-13}$

Comparative Analysis of Root-Finding Methods

Aspect	Bisection Method	False Position (Regula Falsi)	Newton-Raphson	Secant Method
Basic Idea	Bisects interval, selects sub-interval where root lies	Uses straight line between endpoints to approximate root	Uses tangent line: $x_{n+1} = x_n - f(x_n)/f'(x_n)$	Similar to Newton but derivative-free; uses last two points
Initial Requirement	Two guesses with opposite signs	Two guesses with opposite signs	Single guess + derivative	Two initial guesses
Guarantee of Convergence	Always converges	Converges but may stagnate	Not guaranteed	Not guaranteed
Convergence Rate	Linear (slow)	Linear, faster than bisection	Quadratic (fast)	Super-linear
Iterations Needed	High (slow)	Moderate	Low (fast)	Moderate

Comparative Analysis of Root-Finding Methods

Aspect	Bisection Method	False Position (Regula Falsi)	Newton-Raphson	Secant Method
Function Evaluations	1 per iteration	1 per iteration	2 per iteration (f, f')	2 per iteration
Derivative Requirement	Not needed	Not needed	Required	Not required
Robustness	Very robust, guaranteed convergence	Fairly robust, may slow down	Sensitive to guess, may diverge	Less robust than bisection
Use Cases	When guaranteed convergence is important	When faster than bisection is needed	When derivative + good guess available	When no derivative, faster than bisection
Accuracy	Moderate	Moderate, may stall	High, fewer iterations	Good, function-dependent